PyGameMaker — Tutorial 8: Lunar Lander — Answer Key

> For teachers only do not hand out to students. The numbering below matches this tutorial's Worksheet.

Worksheet Answer Key

Part A: 1-C, 2-B, 3-D, 4-A.

Part B: 1. Create (Set gravity). 2. Keyboard: Up Arrow (held). 3. Collision with obj_pad. 4. Collision with obj_ground (with Restart room). 5. Draw (on the controller).

Part C:

1. The vertical speed stays at about -2 and gravity adds only 0.05 each step, so it takes about 40 steps for the climb to stop. That is why the lander keeps rising before it falls.
2. No, the speed does not matter in the tutorial's game. To make a hard landing a crash, test the vertical speed in the pad event: if it is above a limit, crash; otherwise land.
3. Gravity keeps pushing the lander into the pad, so the collision event fires again and again. Switch gravity off (strength 0) in the pad event.

Parts D and E: completion and reflection.